Machine-Learning Surrogate Optimization for Time-Based Consolidation in Last-Mile Parcel Delivery*

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Abstract

Time-based consolidation (TBC) lets logistics service providers delay eligible parcels for batched delivery, trading customer waiting time against operational savings. While the principle is well known, its cost-service balance, its spatial signature, and its fleet-side benefits across heterogeneous carriers remain insufficiently quantified. We optimize TBC schedules on the Hanover Region case (seven LSPs, 1.26 million weekly parcels) using a machine-learning surrogate that couples Daganzo's continuum approximation with a LightGBM residual (2.95% MAPE under postal-code-grouped cross-validation), embedded in a coordinate-descent search over a service-penalty P and a willingness-to-wait share θ. Cost savings reach 22.8% at the cost-optimal extreme, three carrier classes emerge with distinct sweet spots in $P \in [0.25, 0.75]$ € per parcel-day, and the Mo–Sa fleet coefficient of variation drops by up to 60% in that range. Rural postal-code areas with long stems, large service area, and few delivered parcels per stop benefit most. Out-of-sample VROOM re-routing confirms the surrogate is conservative, underestimating realized savings by 1.3–2.1 pp.

Keywords: Temporal consolidation, Parcel delivery, Last-mile logistics, Vehicle routing problem, Delivery scheduling, Willingness to wait

Preprint note

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1. Introduction and Related Work

In the European Union, 78% of internet users purchased goods or services online in 2025 (Eurostat, 2025), illustrating how deeply parcel delivery is embedded in everyday consumption. Each online order creates a physical delivery task in the final segment of the parcel supply chain, commonly referred to as last-mile delivery (LMD), which is among the most expensive, inefficient, and environmentally burdensome parts of the supply chain (Olsson et al., 2019). For logistics service providers (LSPs), this intensifies existing pressures: LMD operations are already

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characterized by low productivity, restrictive delivery time windows, complex urban operating conditions, and high cost shares (Bachofner et al., 2022), reinforced by climate change, rapid urbanization, demographic change, and shifting demand patterns (Gruchmann et al., 2018).

At the same time, e-commerce has made delivery speed a visible part of the customer-facing service promise, so LSPs are expected to provide faster, more personalized, and more reliable services while prices and margins remain constrained (Vivaldini, 2022). However, not all customers prioritize the fastest possible delivery: Nguyen et al. (2019) distinguish price-oriented, convenience-oriented, and value-for-money-oriented segments, indicating that some customers may accept slower delivery if it lowers cost. This heterogeneity makes delivery timing a potential operational control variable. Temporarily storing eligible parcels and dispatching them jointly can reduce costs compared with direct delivery (Peng et al., 2023), and a some-day delivery option can reduce cost and vehicle requirements when customers accept a latest delivery day (Voigt et al., 2025); sustainability benefits, however, arise only if the temporal flexibility translates into better routing, lower fleet requirements, or reduced travel distances (Zeelenberg et al., 2026). Time-based consolidation (TBC) packages this trade-off into a delivery concept in which eligible customers accept a longer delivery window, for example in exchange for a lower price (Naeimian et al., 2026), so that eligible shipments are temporarily held and grouped into batched deliveries within a predefined service promise. TBC therefore moves away from a uniform immediacy logic toward a differentiated service design in which customers retain choice while LSPs gain temporal flexibility.

This paper investigates TBC as an operational response to the tension between customer expectations, LSP constraints, and sustainability objectives. From a planning perspective, the core task is to assign eligible parcels to delivery days at the postal-code-area level such that delivery density increases, additional waiting time remains limited, and weekday demand peaks or uneven fleet use are avoided. Because each schedule induces a different demand pattern and thus different delivery routes, direct evaluation would require solving many VRPs, which is computationally infeasible given the VRP's NP-hardness (Lenstra and Kan, 1981). We therefore propose an ML-based surrogate-optimization framework for identifying cost-efficient TBC schedules in a large-scale case study of the Hanover Region in Germany. The framework combines a postal-code-area scheduling model with a routing-cost surrogate trained on VRP solutions; building on ML-supported combinatorial optimization (Bengio et al., 2021) and data-driven extensions of analytical LMD route-distance approximations (Figliozzi, 2008; Kirubakaran et al., 2026), it enables efficient schedule search across adoption scenarios.

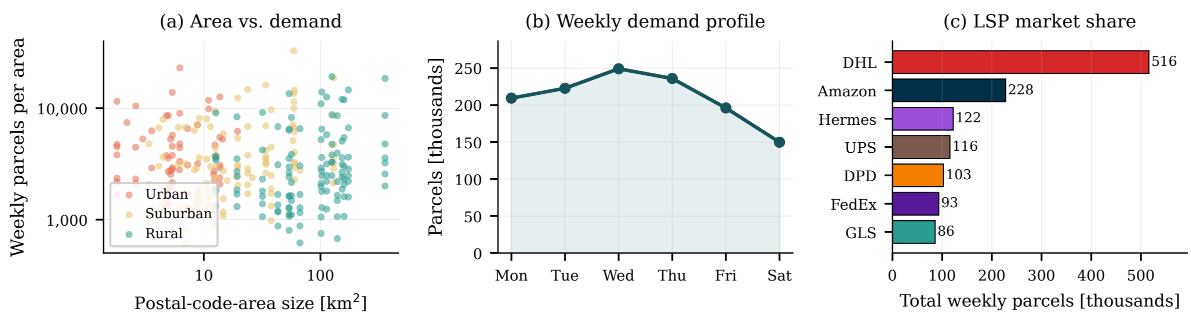

Figure 1: Input data from HAGRID (a) Postal-code-area size vs. weekly volume by region type; (b) weekly demand profile; (c) LSP market share.

2. Methodology

Our framework used weekly parcel-demand patterns generated with the HAGRID model (Bienzeisler et al., 2026), covering seven LSPs in the Hanover Region (about 1.26 million parcels and 1.91 M€ of baseline operating cost per week). As Figure 1 shows, demand varies strongly between dense urban and sparse rural postal-code areas, peaks midweek, and is lowest on Saturday. To make repeated routing evaluations tractable, we decomposed the study area into postal-code-level routing subproblems while preserving the demand, spatial, and depot-access characteristics relevant for delivery cost. The pipeline structure is summarized in Figure 2.[1]

The unit of analysis is a *cell*, defined as an LSP–postal-code-area pair assigned to the provider's serving hub and described by daily demand, area size, stop structure, and depot-access distance. The decision variable is the weekly *delivery pattern*, i.e., the service weekdays for that cell; parcels from non-service days are held until the next scheduled service day. LSP operating costs are route-dependent. Following open-data-based LMD modeling (Hörl et al., 2025), we generated routing-cost labels with VROOM (Coupey et al., 2024) based on Valhalla-derived travel times and distances on OpenStreetMap road geometry.[2] Recurrent congestion was approximated by applying an empirically derived correction factor to free-flow travel times. The current daily-delivery operation served as the baseline for all consolidation scenarios.

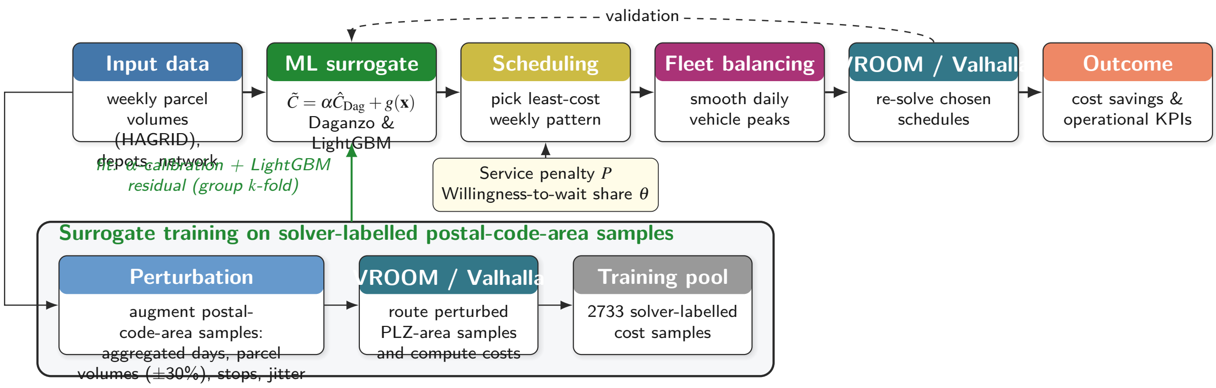

Figure 2: Surrogate-based optimization and validation pipeline for TBC.

2.1. *Cost approximation and machine-learning surrogate*

Solving a full VRP for every candidate schedule is prohibitive, so we approximated operational cost with a continuous model based on Daganzo's distance formulation (Daganzo, 1984) and corrected it with a data-driven term. For area z on day d, route distance and cost are

$$\hat{D}_{z,d} = 2m_{z,d}r_z + \beta\sqrt{n_{z,d}A_z}, \qquad \hat{C}_{z,d} = c_d\hat{D}_{z,d} + c_t\left(\hat{D}_{z,d}/v + sn_{z,d}\right) + c_f m_{z,d}, \tag{1}$$

with $m_{z,d} = \lceil p_{z,d}/Q \rceil$ tours (parcel volume $p_{z,d}$, vehicle capacity Q), average depot distance r_z, $n_{z,d}$ stops over area A_z, network/shape factor β, and distance/time/fixed-cost rates c_d, c_t, c_f, speed v, and per-stop service time s. Consolidation lowers cost by raising delivery density while the capacity term penalizes over-consolidation. Such approximations reproduce the first-order scaling but are point-inaccurate, as they abstract from the road network, depot position, and

[1]The implementation of the proposed framework is available at: https://github.com/TUBS-IVS/vroom-valhalla-lmd-hannover

[2]https://github.com/valhalla/valhalla

demand clustering, normally requiring manual recalibration (Figliozzi, 2008). Instead, we used Equation (1) as a physically interpretable *backbone* and learned its deviation from solver cost,

$$\tilde{C}_{z,d} = \alpha \hat{C}_{z,d} + g(\mathbf{x}_{z,d}), \tag{2}$$

where α is a global calibration factor and the structured residual $g(\cdot)$ is a LightGBM ensemble (Ke et al., 2017), chosen because gradient-boosted trees handle tabular nonlinearities and feature interactions well, and expose feature importance for interpretation (Grinsztajn et al., 2022). The scalar α was calibrated once as the median ratio of solver cost to the raw Daganzo prediction across the training pool, and $g(\cdot)$ then learned the structured remainder.

The feature vector $\mathbf{x}_{z,d}$ comprised 25 demand and geometry descriptors derived ex ante from the data, following the taxonomy of Akkerman and Mes (2025) (parcel volume, stops, service-area size, depot distance, parcels per stop, the load factor $p_{z,d}/Q$, and per-stop demand statistics), with VROOM/Valhalla route costs as training targets. To cover the consolidation cost range rather than only baseline operations, we augmented the pool with synthetic cell-day samples obtained by demand scaling, stop dropout, coordinate jitter, and multi-day aggregation, the last reproducing the elevated per-day volumes of batched delivery. Generalization to both sparse rural and dense urban areas was assessed by group k-fold cross-validation with folds grouped by postal-code area.

2.2. *Schedule optimization and fleet balancing*

For each *cell* (each LSP, postal-code area pair), we enumerated all weekly delivery patterns feasible under a maximum holding time of $H_{\max} = 3$ days over the six-day operating week, yielding 39 admissible patterns. For a pattern σ, a delivery day served its own demand plus the parcels held from the preceding non-delivery days, while non-delivery days carried no consolidated tour. The average customer waiting time $\bar{w}(\sigma)$ is the mean delay over the six weekdays between parcel arrival and the next scheduled service day under a cyclic interpretation of σ. Cells served by the same depot share a pooled daily delivery tour for the non-consolidated share of their demand, so the surrogate cost of a cell is not separable from the patterns chosen by its hub neighbors. We therefore solved the system-wide selection by coordinate descent: at every update the focal cell chose the pattern that minimized the weekly surrogate cost evaluated against the current schedules of all other cells at the same hub, augmented by a linear service penalty,

$$\sigma_z^\star = \arg\min_{\sigma \in \mathscr{S}} \left[\sum_d \tilde{C}_{z,d}(\sigma \mid \sigma_{-z}) + P\,\theta_z\, p_z\, \bar{w}(\sigma) \right], \tag{3}$$

where $\mathscr{S}$ is the set of feasible patterns, σ_{-z} the fixed patterns of all other cells at the focal cell's hub, p_z the weekly parcel volume, and θ_z the share of parcels eligible for consolidation in the cell. Following a business-prioritized power law of the system-wide adoption θ, business recipients opt in earlier than private ones, so each cell's θ_z reflects its own composition while the volume-weighted regional average equals θ; only this willing fraction is consolidated, the rest delivered conventionally. Coordinate descent iterates with multiple random restarts until no single-cell update improves the system-wide objective. The service penalty P is a shadow price (Boyd and Vandenberghe, 2004) of waiting, i.e. the marginal cost the operator trades for one additional parcel-day of delay, and enters (3) as a steering term only, not as a cost component, which keeps the cost-service trade-off continuous across the grid. We swept (P, θ) over eight P levels from 0 to 10 €/p/d and eleven θ levels from 0 to 1 in steps of 0.1.

A cost-optimal selection alone can concentrate deliveries on a few weekdays, producing peaks that are hard to staff under realistic subcontractor and roster constraints. A two-stage post-processing therefore preserves each cell's chosen delivery frequency: a per-depot greedy local search reassigns weekdays to flatten each depot's peak, accepting only swaps that keep total weekly cost within 5% of the cost-optimal value, and a subsequent system-level smoothing stage redistributes weekday assignments across hubs to flatten the Monday-to-Saturday system fleet.

3. Results and Discussion

3.1. Surrogate model: quality and validation

Under group k-fold cross-validation with folds grouped by postal-code area, so that no area appeared in both training and validation, the Daganzo-LGB-Hybrid achieved 2.95% MAPE ($R^2 = 0.997$). We kept the Daganzo cost parameters at standard operational values and calibrated the global cost level through the factor α, defined as the median ratio of solver cost to Daganzo cost ($\alpha = 1.34$). Figure 3 reports the out-of-fold accuracy in three stages: the uncalibrated approximation underestimated cost by about 26% (panel a); the global α calibration removed this systematic bias and reached 9.7% MAPE (panel b); and the learned LightGBM residual captured remaining effects related to capacity thresholds, stop density, and depot distance, reducing the error to 2.95% with near-zero bias (panel c). Overall, among the benchmark models tested in this study, including both statistical and ML approaches, the Daganzo-LGB-Hybrid achieved the best predictive performance.

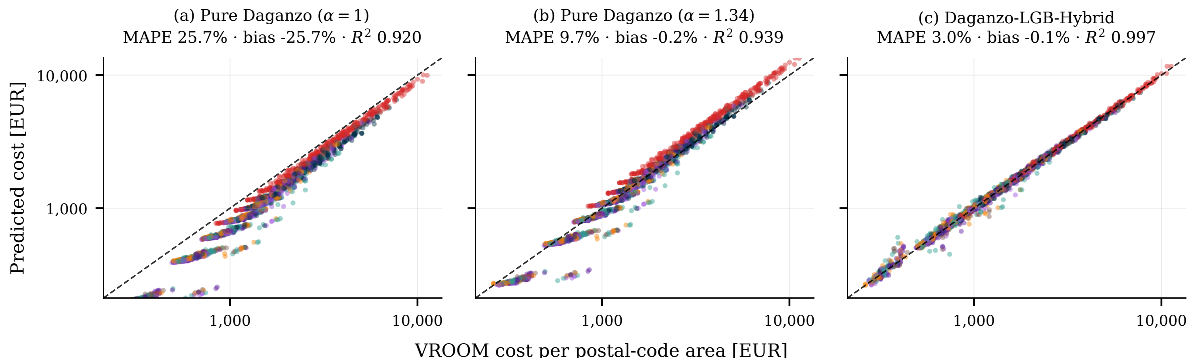

Figure 3: Predicted vs. VROOM routing cost under group k-fold validation by postal-code area. Points are coloured by LSP. Panels show (a) the raw Daganzo approximation, (b) the globally α-calibrated Daganzo approximation, and (c) the final Daganzo-LGB-Hybrid surrogate.

3.2. Schedule optimization and the cost–service trade-off

Figure 4 traces the optimized delivery-day distribution across the 312 postal-code areas as θ rises; we summarize each scenario by the mean delivery frequency $\bar{f}$, the unweighted average number of weekly delivery days across cells. At $\theta = 0$ every area is served daily at any penalty, recovering the baseline. As θ rises, P sets the pace: at $(P,\theta) = (0,1)$, 97.4% of areas adopt the minimum two-day pattern ($\bar{f} = 2.03$); the remaining 2.6% are high-volume urban cells whose two-day demand would breach vehicle capacity, so the Daganzo capacity term forces an extra day. The mode shifts upward through $P = 0.5$ ($\bar{f} = 3.95$, 24% daily at the capacity floor) and $P = 1$ ($\bar{f} = 4.93$, 54% daily); for $P \geq 5$ and $\theta \geq 0.3$ the system reverts to fully daily delivery. This reversion is non-monotone in θ: at $\theta = 0.1$ even $P = 10$ still leaves 42% of areas non-daily ($\bar{f} = 5.05$), with the daily share rising only from $\sim 14\%$ at $P \leq 1$ to 58% at $P = 10$. Two mechanisms compound. First, the penalty scales as $P \cdot \theta_z \cdot p_z \cdot \bar{w}$, so the effective pressure $P \cdot \theta$ at $(10, 0.1)$ matches that at $(1,1)$ (1 € per parcel-day). Second, hub-bundling lets rural and

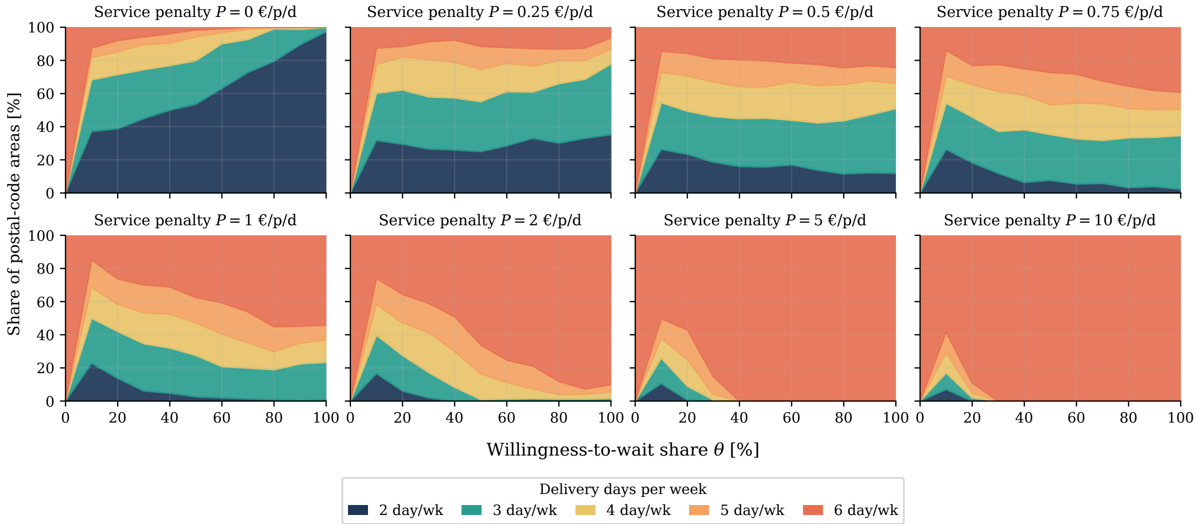

Figure 4: Delivery-frequency mix across postal-code areas as a function of the willingness-to-wait share θ at different service penalties P.

low-density suburban cells share a depot tour with neighbors, so the routing saving on the 90% standard-delivery parcels covers the small penalty on the willing 10%. The balance is fragile because θ enters linearly: at $\theta = 0.2$ only 11% remain non-daily at $P = 10$, and by $\theta = 0.3$ $(P \cdot \theta = 3)$ the offset is exhausted.

Figure 5 reports cost and fleet outcomes across the (P, θ) grid. Cost saving relative to the daily-delivery baseline (a, b) peaks at 22.8% in the fully consolidated regime $(P, \theta) = (0, 1)$ and decays monotonically with P; fleet balancing costs only 0.3–0.6 pp of saving in the high-saving region, so schedule selection captures the bulk of the benefit while balancing trades a marginal cost increment for fleet smoothness. The customer-side cost grows much more slowly than the saving decays: the parcel-weighted additional wait drops from 0.98 d at $P = 0$ to 0.45 d at $P = 0.25$ and to 0.23 d at $P = 0.5$ (c), more than halving with just 4.2 pp of saving lost over $[0, 0.25]$ before the marginal cost of further wait reduction triples from 7.9 to 23.2 pp/d over $[0.25, 0.5]$. On the fleet side, the Mo–Sa coefficient of variation falls by 54% at $(0.5, 1)$ and up to 71% at the most consolidated points (e), and peak fleet by 12.9% at $(0.5, 1)$ with 8–14% across the cost-friendly region (d), but the total weekly fleet (f) only drops meaningfully at full adoption (-10% at $P = 0$, -8.3% at $P = 0.25$, -5.9% at $P = 0.5$) and can grow by up to 4.6% at intermediate θ. The reason is structural: as long as some parcels remain in conventional delivery, the operator must run a daily conventional tour in parallel with the batched delivery days, so the peak and CV gains of (d) and (e) reflect a redistribution of vehicle requirement across weekdays rather than an absolute reduction.

The per-LSP operating point $P^\star$ is identified as the chord-distance knee of each carrier's (saving, wait) Pareto front at $\theta = 1$ (Figure 6, a), i.e. the point with the largest orthogonal distance to the chord between the cost-minimum and wait-minimum extremes, beyond which the saving lost per further day of wait reduction grows sharply. This sorts the seven operators into service-bound (Amazon, DHL) at $P^\star = 0.25$, hybrid (FedEx, Hermes, UPS) at $P^\star = 0.5$, and cost-aggressive (DPD, GLS) at $P^\star = 0.75$. Which areas can consolidate is governed by the local routing geometry: the per-area saving correlates strongly with parcels per drop-site ($\rho = -0.72$), hub distance ($\rho = +0.53$), and area size ($\rho = +0.31$); dense urban cells with short stems and high parcels-per-stop gain little from waiting, while sparse rural cells with long stems benefit most. The same mechanism caps the service-bound class at a 10% per-postal-code

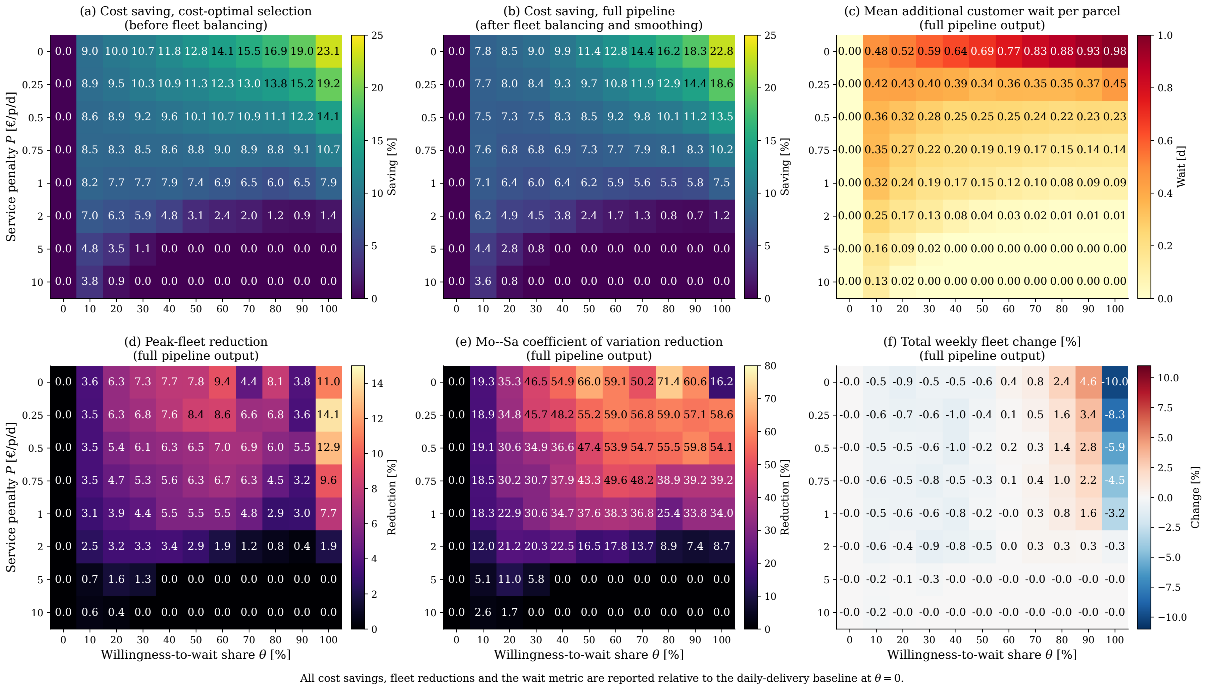

Figure 5: Total cost saving (a, b), parcels-weighted additional customer wait (c), and fleet metrics (d-f) over the service-penalty P and willingness-to-wait θ grid, reported relative to the daily-delivery baseline.

median saving while the hybrid and cost-aggressive classes reach 22–24% at their $P^{\star}$ (b), and at low adoption ($\theta \leq 0.3$) it even pushes 35–44% of service-bound cells into negative saving, since the parallel conventional tour for the 70–90% non-willing parcels exceeds the batching gain for cells near the capacity floor. At each $P^{\star}$ and full adoption, rural areas save a median of 25% versus 9% for urban areas (c), and the structural splits (d–f) confirm this ordering: median saving rises from 8% to 28% with hub distance and from 10% to 25% with area size, while it falls from 27% to 4% with parcels per drop-site between quartiles. TBC therefore pays most where delivery is sparse and far from the depot. This efficiency pattern, however, creates an equity concern: service quality between urban and rural areas should be maintained (Pereira et al., 2017), especially since rural and suburban deliveries account for a substantial share of LMD activity (Bienzeisler et al., 2026).

As a final step, we re-routed the optimized schedules with VROOM as an out-of-sample validation of the surrogate. At $\theta = 1$ and the three validated operating points $P \in \{0, 0.25, 0.5\}$, the Daganzo-LGB-Hybrid matches the solver-actual system-wide cost saving within 1.3–2.1 pp: 22.8% predicted vs. 24.3% actual at $P = 0$, 18.6% vs. 19.8% at $P = 0.25$, and 13.5% vs. 15.6% at $P = 0.5$. The surrogate is consistently conservative, slightly understating the achievable savings across the scenario space.

4. Conclusion

We presented an ML-based surrogate optimization framework for time-based consolidation in last-mile parcel delivery on a case study of seven LSPs, 312 postal-code areas, and 1.26 million parcels per week in the Hanover Region. The Daganzo-LightGBM hybrid surrogate achieved 2.95% MAPE under postal-code-grouped cross-validation, enabling a full (P, θ) sweep. The operationally efficient range lies between $P = 0.25$ and $P = 0.5$ €/p/d, capturing up to 22.8% cost saving at $\theta = 1$ and a 54% reduction in the Mo–Sa fleet coefficient of variation, while the absolute weekly vehicle count only falls at full adoption. Per-LSP chord-distance

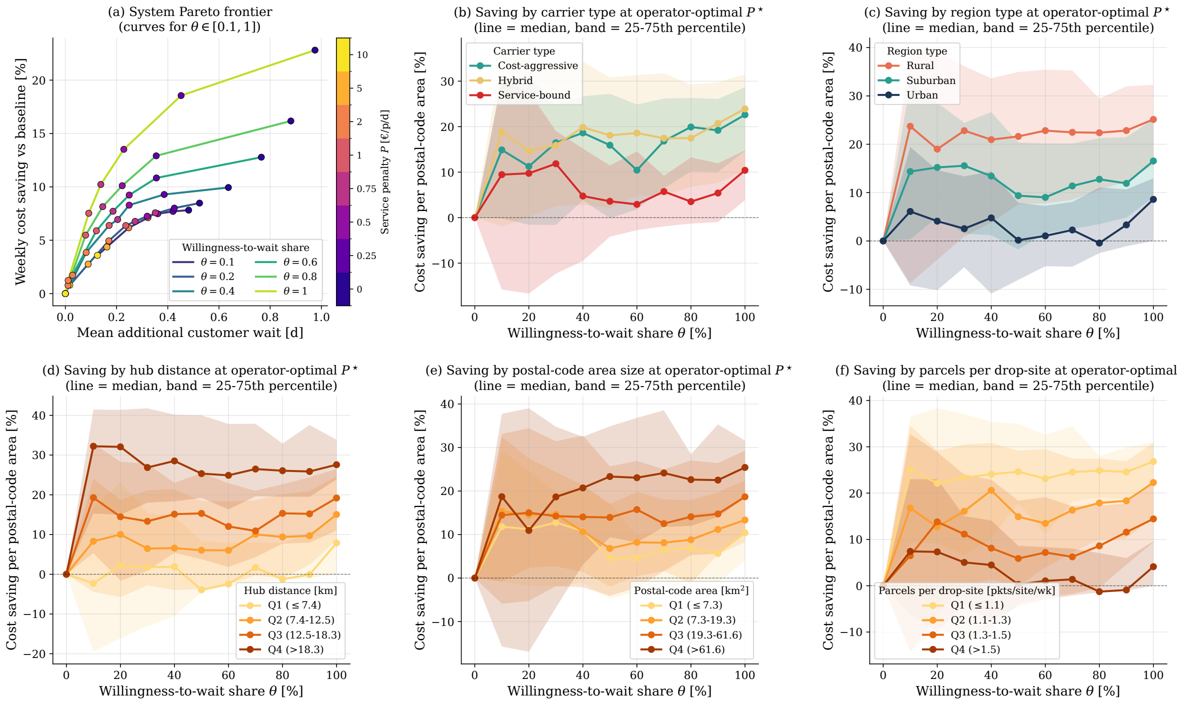

Figure 6: System Pareto frontier (a) and median per-postal-code cost saving by carrier type (b), region type (c), hub distance (d), postal-code area (e), and parcels per drop-site (f) over the willingness-to-wait share θ at each LSP's optimal service penalty $P^\star$.

optima sort the operators into service-bound, hybrid, and cost-aggressive classes, with savings driven by long stems, large postal-code areas, and few parcels per drop-site. TBC therefore emerges as a particularly attractive lever for rural regions, where the structural gains are largest. Validation relies on VROOM as a routing benchmark rather than ground truth, and our per-postal-code-area decomposition rules out inter-area consolidation that could further amplify TBC gains. Future work will lift this constraint by jointly routing neighbouring postal-code areas within the solver's tractability limits.

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